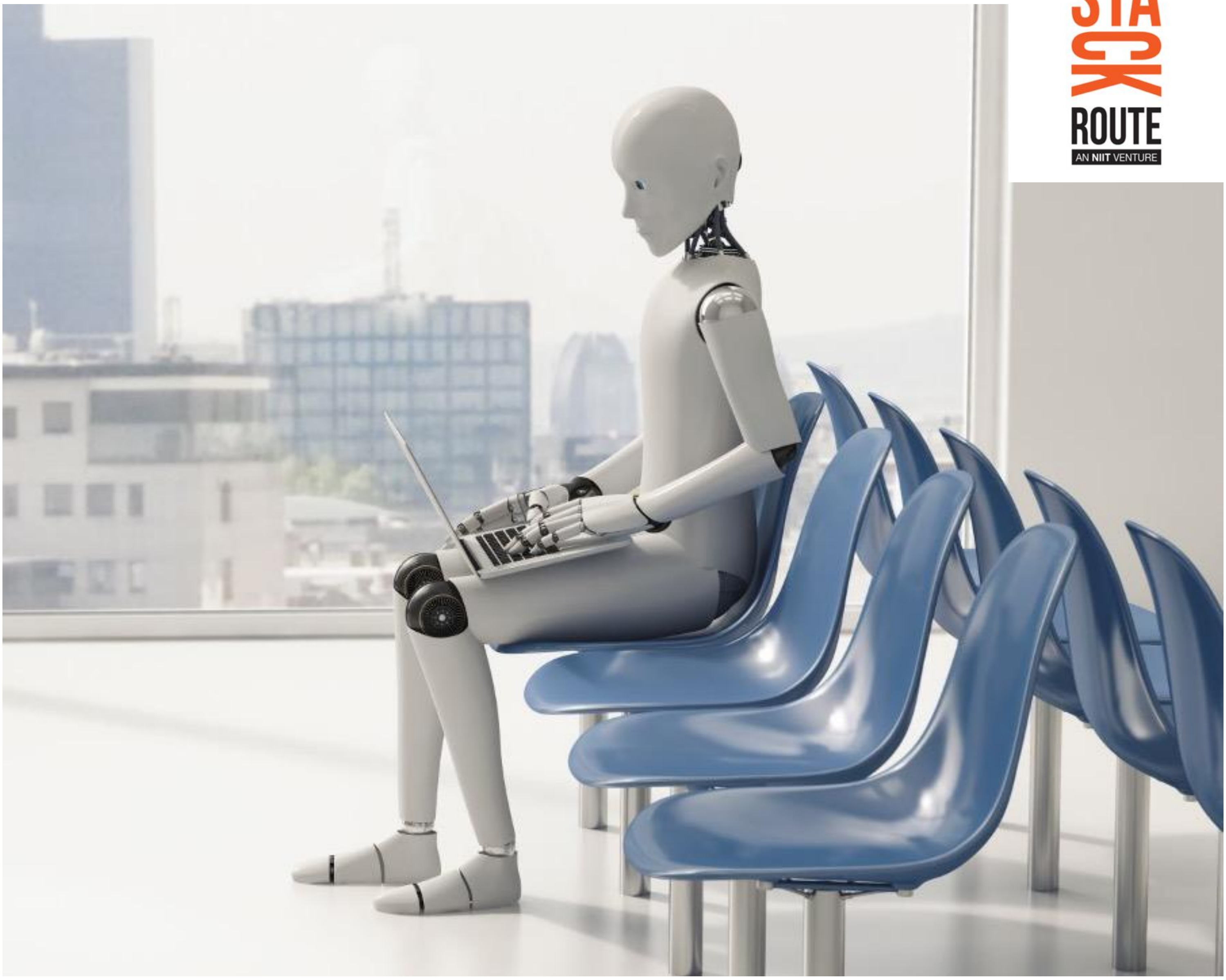

Assignment: Banking Data Pipeline using Medallion Architecture and Dataflow

Build a Banking Analytics Pipeline using Medallion Architecture and Dataflow

1. Business Scenario

A retail bank wants to modernize its data platform to support better reporting on customers, accounts, transactions, loans, and branch-level performance.

Currently, data is coming from multiple CSV files maintained by different banking teams. The data has duplicate records, missing values, inconsistent formats, invalid transaction amounts, and delayed entries.

The bank wants participants to design a simple **Bronze → Silver → Gold** pipeline using **Medallion Architecture** and **Dataflow Gen2**.

The final output should help business users answer questions such as:

1. **Which branches have the highest transaction volume?**
2. **Which customer segment contributes the highest account balance?**
3. **How many transactions are successful, failed, or pending?**
4. **Which loan types have the highest outstanding amount?**
5. **Which customers show high transaction activity?**

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2. Dataset Overview

Participants will work with the following banking datasets:

Dataset	Description
`customers.csv`	Customer demographic and segment details
`accounts.csv`	Bank account details linked to customers
`transactions.csv`	Transaction-level banking data
`loans.csv`	Loan details for customers
`branches.csv`	Branch and region details

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3. Sample Dataset Structure

A. customers.csv

Column	Description
customer_id	Unique customer ID
customer_name	Name of the customer
gender	Male/Female/Other
age	Customer age
city	Customer city
customer_segment	Retail, Premium, SME, Corporate
onboarding_date	Date when customer joined the bank
kyc_status	Verified, Pending, Rejected

B. accounts.csv

Column	Description
account_id	Unique account ID
customer_id	Linked customer ID
branch_id	Linked branch ID
account_type	Savings, Current, Salary, NRI
account_open_date	Account opening date
current_balance	Current account balance
account_status	Active, Dormant, Closed

C. transactions.csv

Column	Description
transaction_id	Unique transaction ID
account_id	Linked account ID
transaction_date	Date of transaction
transaction_type	Debit, Credit, UPI, ATM, NEFT, IMPS
transaction_amount	Amount of transaction
transaction_status	Success, Failed, Pending
channel	Mobile Banking, Internet Banking, ATM, Branch

D. loans.csv

Column	Description
loan_id	Unique loan ID
customer_id	Linked customer ID
loan_type	Home Loan, Personal Loan, Auto Loan, Business Loan
loan_amount	Total approved loan amount
outstanding_amount	Remaining loan amount
interest_rate	Loan interest rate
loan_status	Active, Closed, Defaulted

E. branches.csv

Column	Description
branch_id	Unique branch ID
branch_name	Name of branch
city	Branch city
region	North, South, East, West, Central
branch_manager	Branch manager name

4. Assignment Tasks

Task 1: Create the Bronze Layer

Objective

Ingest raw banking CSV files into the Lakehouse without applying heavy transformations.

Participant Activities

1. **Upload all raw CSV files into the Lakehouse Files section.**
2. **Create Bronze tables for each dataset:**
   - `bronze_customers`
   - `bronze_accounts`
   - `bronze_transactions`
   - `bronze_loans`
   - `bronze_branches`
3. **Add ingestion metadata columns:**
   - `source_file_name`
   - `ingestion_timestamp`
   - `pipeline_run_id`
4. **Preserve the raw structure of the data.**

Expected Output

Bronze Table	Expected Result
`bronze_customers`	Raw customer data loaded
`bronze_accounts`	Raw account data loaded
`bronze_transactions`	Raw transaction data loaded
`bronze_loans`	Raw loan data loaded
`bronze_branches`	Raw branch data loaded

Task 2: Perform Data Profiling using Dataflow Gen2

Objective

Use Dataflow Gen2 to inspect data quality issues before moving to the Silver layer.

Participant Activities

1. **Create a Dataflow Gen2 for banking data profiling.**
2. **Connect to Bronze tables or raw files.**
3. **Identify the following data issues:**
   - Duplicate customer IDs
   - Missing customer names
   - Invalid age values
   - Missing or negative transaction amounts
   - Inconsistent transaction statuses
   - Null account balances
   - Invalid loan status values
   - Duplicate transaction IDs
4. **Create a profiling summary table.**

Expected Output Table: `banking_data_quality_summary`

Dataset	Total Records	Duplicate Records	Missing Values	Invalid Records	Remarks
Customers					
Accounts					
Transactions					
Loans					
Branches					

Task 3: Create the Silver Layer

Objective

Clean, validate, standardize, and prepare conformed banking datasets.

Participant Activities

Create the following Silver tables:

1. `silver_customers`
2. `silver_accounts`
3. `silver_transactions`
4. `silver_loans`
5. `silver_branches`

Required Transformations

Dataset	Transformation Requirements
Customers	Remove duplicate `customer_id`, handle missing names, standardize gender and segment values, validate age
Accounts	Remove duplicate `account_id`, replace null balances with 0, standardize account status, validate branch ID
Transactions	Remove duplicate `transaction_id`, remove negative or null transaction amounts, standardize transaction status and channel
Loans	Validate loan amount and outstanding amount, standardize loan status, check interest rate range
Branches	Remove duplicate branch records, standardize region and city names

Example Silver Rules

Rule ID	Rule Description
R1	Customer age must be between 18 and 100
R2	Transaction amount must be greater than 0
R3	Account balance should not be null
R4	KYC status should be Verified, Pending, or Rejected
R5	Loan status should be Active, Closed, or Defaulted
R6	Transaction status should be Success, Failed, or Pending

Expected Output

Cleaned and standardized Silver tables ready for business-level aggregation.

Task 4: Create the Gold Layer

Objective

Build business-ready analytical tables for banking reporting.

Participant Activities

Create the following Gold tables:

1. `gold_branch_transaction_summary`

Column	Description
branch_id	Branch ID
branch_name	Branch name
city	Branch city
region	Branch region
total_transactions	Total number of transactions
total_transaction_amount	Total transaction value
successful_transactions	Count of successful transactions
failed_transactions	Count of failed transactions

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2. `gold_customer_value_summary`

Column	Description
customer_id	Customer ID
customer_name	Customer name
customer_segment	Customer segment
total_accounts	Number of accounts
total_balance	Total balance across accounts
total_transactions	Number of transactions
total_transaction_amount	Total transaction value
loan_amount	Total loan amount
outstanding_amount	Total outstanding amount

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3. `gold_loan_portfolio_summary`

Column	Description
loan_type	Type of loan
total_customers	Number of customers with loans
total_loan_amount	Total approved loan amount
total_outstanding_amount	Total outstanding amount

Column	Description
defaulted_loans	Count of defaulted loans

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4. `gold_channel_performance_summary`

Column	Description
channel	Banking channel
total_transactions	Total transaction count
total_transaction_amount	Total transaction value
success_rate	Percentage of successful transactions
failed_rate	Percentage of failed transactions

Task 5: Build Dataflow Gen2 Pipeline

Objective

Use Dataflow Gen2 to create an automated transformation flow from Bronze to Silver or Silver to Gold.

Participant Activities

1. **Create one Dataflow Gen2 for cleaning customer and account data.**
2. **Create one Dataflow Gen2 for transaction standardization.**
3. **Create one Dataflow Gen2 for Gold-level business summary tables.**
4. **Load the output into Lakehouse or Warehouse tables.**
5. **Document each transformation step.**

Required Dataflow Steps

Step	Activity
Source Connection	Connect to Lakehouse Bronze tables
Data Type Conversion	Correct date, numeric, and text columns
Remove Duplicates	Remove duplicate customer, account, and transaction records
Replace Nulls	Replace null balances and missing values
Filter Invalid Rows	Remove invalid transaction and loan records
Merge Queries	Join customers, accounts, transactions, loans, and branches
Group By	Create business-level summaries
Destination	Load output into Silver or Gold tables

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5. Business Questions to Answer

Participants should use the Gold layer to answer the following:

1. **Which branch has the highest transaction value?**
2. **Which region has the highest number of successful transactions?**
3. **Which customer segment has the highest total account balance?**
4. **What percentage of transactions failed across each channel?**
5. **Which loan type has the highest outstanding amount?**

6. **Which customers have both high transaction volume and active loans?**
7. **Which branches have more failed transactions than the average failed transaction count?**
8. **What is the overall success rate of digital banking channels?**

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6. Deliverables

Participants must submit:

Deliverable	Description
Bronze Tables	Raw ingested tables with metadata columns
Data Profiling Summary	Table showing data quality issues
Silver Tables	Cleaned and validated tables
Gold Tables	Business-ready aggregated tables
Dataflow Screenshots	Screenshots showing transformation steps
SQL / Power Query Logic	Queries or transformation steps used
Final Insights Summary	5–7 banking business insights
Architecture Diagram	Bronze → Silver → Gold pipeline flow

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7. Suggested Architecture Diagram

```
Raw Banking CSV Files
        |
        v
Bronze Layer
Raw tables with metadata
        |
        v
Dataflow Gen2 Profiling
Data quality checks and issue summary
        |
        v
Silver Layer
Cleaned, validated, standardized banking data
        |
        v
Gold Layer
Business-ready banking analytics tables
        |
        v
Power BI / SQL Reports
Branch, customer, transaction, and loan insights
```

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8. Evaluation Rubric

Criteria	Weightage
Correct Bronze ingestion with metadata	15%
Data profiling and issue identification	15%
Silver layer cleaning and validation logic	25%
Correct Gold layer aggregations	25%
Use of Dataflow Gen2 transformation steps	10%

Criteria	Weightage
Final business insights and documentation	10%

9. Bonus Challenge

Participants can attempt the following for additional marks:

1. **Add incremental load logic using transaction date.**
2. **Handle late-arriving transactions.**
3. **Create a rejected records table for invalid rows.**
4. **Build a Power BI dashboard using Gold tables.**
5. **Add a data quality score for each dataset.**

10. Final Submission Format

Participants should submit one ZIP file containing:

```
banking_medallion_assignment/
│
├── raw_data/
│   ├── customers.csv
│   ├── accounts.csv
│   ├── transactions.csv
│   ├── loans.csv
│   └── branches.csv
│
├── sql_scripts/
│   ├── bronze_validation.sql
│   ├── silver_transformation.sql
│   └── gold_aggregation.sql
│
├── screenshots/
│   ├── dataflow_customer_cleaning.png
│   ├── dataflow_transaction_cleaning.png
│   └── gold_table_output.png
│
├── documentation/
│   ├── architecture_diagram.png
│   └── final_insights_summary.docx
│
└── README.md
```